RESEARCH ~~PORPOSAL~~ DESCRIPTION

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A3 - Next Generation of Exposure and Seismic Risk Maps for the Trentino region

Abstract: Nowadays, the correct assessment of seismic risk has fundamental importance, as it has the dual purpose of preserving people's safety (i.e., emergency management) and protecting the integrity of Italian building heritage (whether modern or historical). Research in this field therefore aims at developing the first Italian prototype of high-resolution seismic exposure and risk maps for Northern Italy, with a local-level detail that is unprecedented at the national level. The scientific context in which this contribution is framed is well represented by the recent publication of the Swiss seismic risk model by the Swiss Seismological Service (SED) (Wiemer, et al. 2023), which confirms the importance of advanced tools for such in-depth analysis.

~~The project results from the collaboration between the National Institute of Oceanography and Experimental Geophysics (OGS), the Seismic and Geological Studies Service of the Autonomous Province of Trento (PAT), the University of Trieste (UniTs), and the Department of Civil Engineering of the University of Trento (UniTn).~~ The main objectives of the ~~initiative, consistent with the timeline outlined in the project GANTT,~~ include the development of innovative methodologies for the estimation of seismic risk at the local scale, the validation of models through historical and instrumental data, and the production of maps that can be immediately useful to authorities in the field of risk mitigation and territorial management.

Research description: Over the past 30 years, numerous studies have been conducted on seismic risk assessment, especially with reference to central Italy, which has been most affected by recent earthquakes. The most destructive among these occurred in L'Aquila (2009) in Abruzzo and in Norcia (2017) in Umbria. The high seismic hazard that characterizes Italian territory, combined with the significant vulnerability of its historical and architectural heritage (at least until the early 2000s, when the first guidelines on seismic action were issued), has made it necessary to develop national (Borzi, Faravelli, et al. 2023), (Zanini, et al. 2019), or regional (Borzi, Onida, et al. 2021) risk mitigation maps and strategies. Additionally, interactive maps for the rapid assessment of damage scenarios following potentially destructive seismic events (Parolai, et al. 2021) have been created to optimize emergency management by civil protection authorities.

A specific risk assessment requires the detailed analysis of all three components that define it. Therefore, the research project is structured into three main phases aimed at building an integrated local-scale model:

1. **WP#1:** The first phase involves the development of seismic microzonation models, essential for estimating the hazard level of the territory based on geological and geomorphological characteristics. The result consists in a unified digital system for collecting and managing information from past and ongoing microzonation campaigns conducted by PAT and OGS (Laurenzano, Garbin e Parolai 2022);
2. **WP#2:** The second phase focuses on the creation of vulnerability and exposure models of the built environment, using advanced machine learning and data mining techniques to analyze existing building stock, combined with the most modern photogrammetric survey technologies.

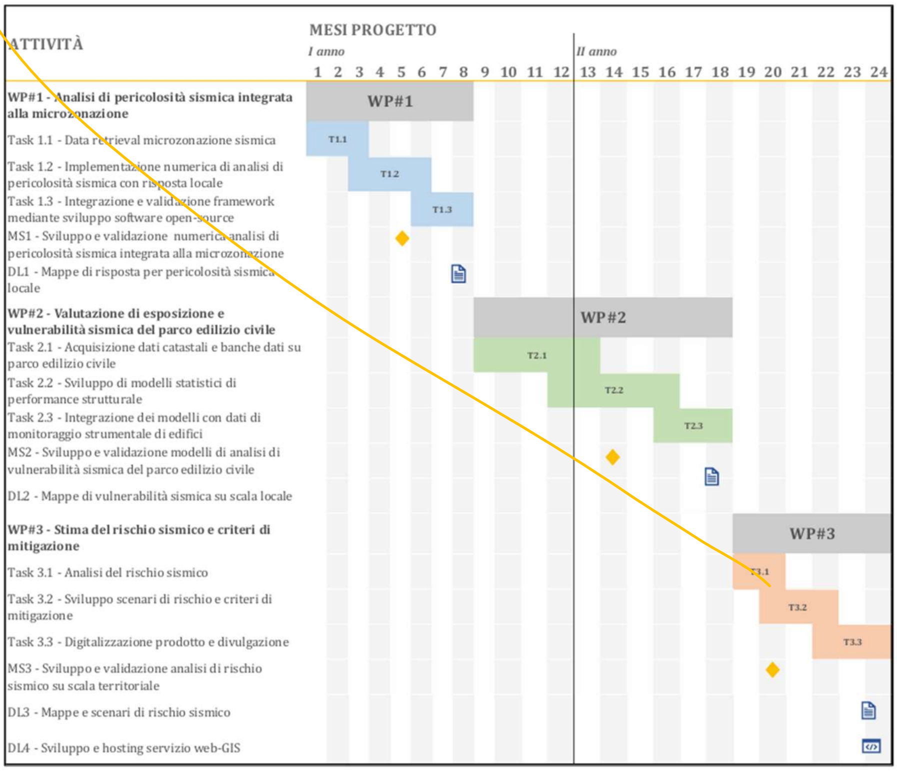
The goal is to produce a map of the seismic vulnerability and exposure of the built heritage in the analyzed areas, harmonized within the digital system previously developed;

3. **WP#3:** Finally, the third phase integrates the results of the previous phases into a software platform for local seismic risk assessment, combining hazard data with structural vulnerability information, in order to generate the first local seismic risk map for the project's pilot areas.

In detail, the first phase of the project (WP#1) focuses on the development of seismic hazard maps, integrated with existing and newly acquired microzonation data for areas of interest in the Trentino region. Specifically, the task "Data retrieval for seismic microzonation" involves using the results of site response analyses carried out collaboratively by PAT, OGS, and the research team led by Professor S. Parolai (University of Trieste). This study provides the spatial distribution of site resonance frequencies, shear wave velocities at ground level, spectral amplification functions and curves, and zonation maps derived through clustering techniques. The experimental data used include ambient noise measurements and seismic recordings, processed using well-established methodologies such as the horizontal-to-vertical spectral ratio of ambient seismic noise (NHV), the Generalized Inversion Technique (GIT), and the horizontal-to-vertical spectral ratio from earthquake recordings (EHV). A temporary network of seismometric stations enabled the estimation of local seismic response, generating amplification functions processed using GIT, EHV, and NHV techniques. The resulting outputs – including resonance frequency maps and spectral amplification curves – are essential inputs for accurate hazard assessments and will be used to define territory-specific Ground Motion Prediction Equations (GMPEs) and implement seismic hazard analyses on the OpenQuake platform. These analyses are based on scenario simulations conditioned by local macroseismic and accelerometric data, and generate ground motion fields from which intensity measures necessary for subsequent risk assessments are extracted.

Subsequently, the assessment of exposure and seismic vulnerability of the civil building stock in the microzonation areas represents the second core element for producing territorial-scale seismic risk analyses (WP#2). The process begins with the collection of data and information from databases and catalogs developed for the North-East of Italy, thanks to the collaboration between OGS and Civil Protection, mainly regarding the structural behavior of buildings located in the areas analyzed. The data, gathered from various sources including land registries, open-access databases (e.g., FRIBAS), APIs from digital platforms (OpenStreetMap, Google Maps), and existing accelerometric networks, are then processed and harmonized. A subsequent phase of data mining and exploratory analysis is carried out to identify the most significant structural features, particularly the fundamental periods (T1) of the buildings. Using advanced machine learning algorithms, clustering techniques are applied to identify homogeneous building classes. For each class, dynamic response models (Single Degree of Freedom – SDOF – or more complex where needed) are developed. This is followed by uncertainty characterization and the initial probabilistic assignment of vulnerability models, based on a Bayesian approach, which can later be updated with structural monitoring data. For critical buildings, such as schools and hospitals, fragility curves are specifically developed using advanced methods such as Incremental Dynamic Analysis (IDA). Finally, initial exposure maps are produced, based on the distribution of fundamental frequencies of the building stock, as illustrated in Figure 4. These maps are compared with the hazard maps from WP#1 to identify potential resonance effects. In parallel, a Bayesian calibration and inversion process is undertaken to update the vulnerability models for each building class. Targeted data collection is also planned in the areas and classes most susceptible to resonance phenomena, with the aim of reducing uncertainty and increasing the reliability of the analyses.

Finally, Work Package 3 (WP#3) focuses on the assessment of local seismic risk, achieved through the integration of the results from the seismic microzonation campaigns conducted by PAT with the exposure and vulnerability data of the civil building stock developed during the project. Based on these results, risk mitigation strategies and operational guidelines are proposed, supporting both emergency management and regular territorial planning activities. The first step consists of integrating the hazard and vulnerability analyses carried out in WP#1 and WP#2, with the objective of producing the first local-scale seismic risk map. This is accomplished by developing specific analysis procedures and workflows in Python, executed through the OpenQuake platform engine. Subsequently, historical earthquake scenarios are simulated to calibrate the risk models and generate reliable damage forecasts. These results offer essential input for defining intervention strategies aimed at both risk reduction and emergency preparedness. The final outcome consists in the development and validation of an integrated digital system for the collection, management, and visualization of data from microzonation studies, structural vulnerability and exposure assessments, as well as structural monitoring measurements. This system will be made accessible through an interactive web-GIS platform, featuring thematic layers and dynamic infographics that display the various maps produced (hazard, vulnerability, risk). Two access modes are planned: one public, for citizens, and one restricted, for technical experts, who will be able to consult and download technical datasets.



The main objectives of the project can be summarized as follows:

1. **Develop a high-resolution seismic hazard map** by integrating site-specific ground motion amplification effects derived from experimental data (seismic noise and earthquake records) using consolidated techniques (NHV, GIT, EHV);

2. **Assess the vulnerability and exposure of the civil building stock** on a territorial scale through structural models (SDOF) and large-scale machine learning techniques, or specific fragility models for strategic buildings;

3. **Produce the first integrated local seismic risk map** by combining hazard maps (site effects) with structural vulnerability maps;

4. **Implement an interactive web-GIS service** for public and technical consultation of hazard, vulnerability, and risk maps, integrating data from microzonation, exposure, and structural monitoring;
5. **Propose a scalable and replicable methodology and workflow** for seismic risk assessment in urban areas and strategic infrastructure at the national level;
6. **Provide operational tools for territorial planning**, risk mitigation, and emergency management.

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